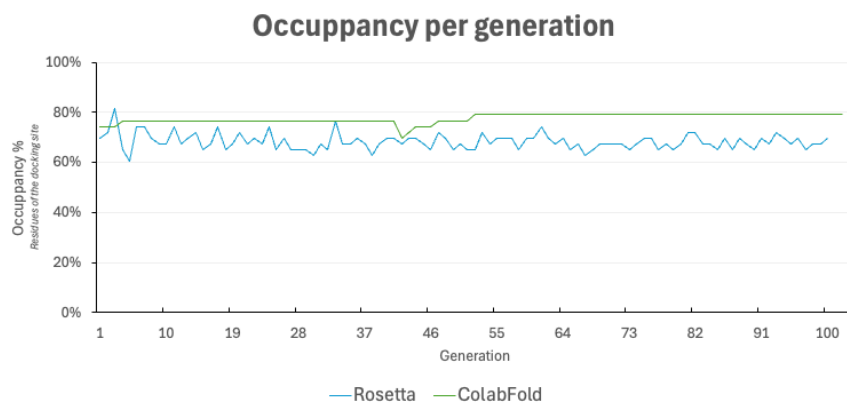
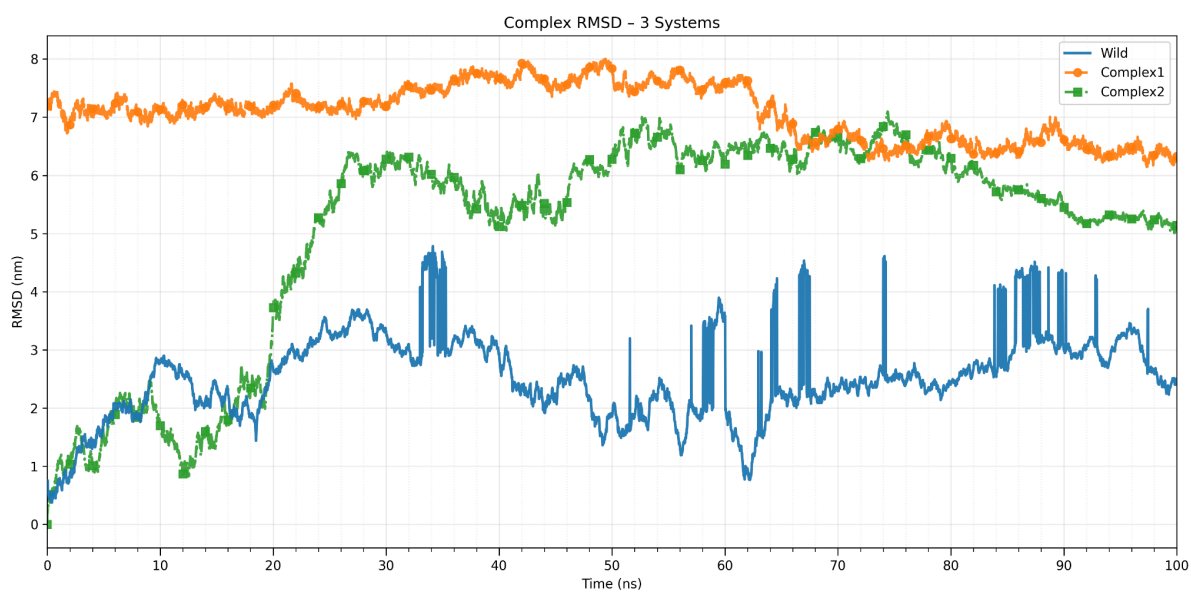


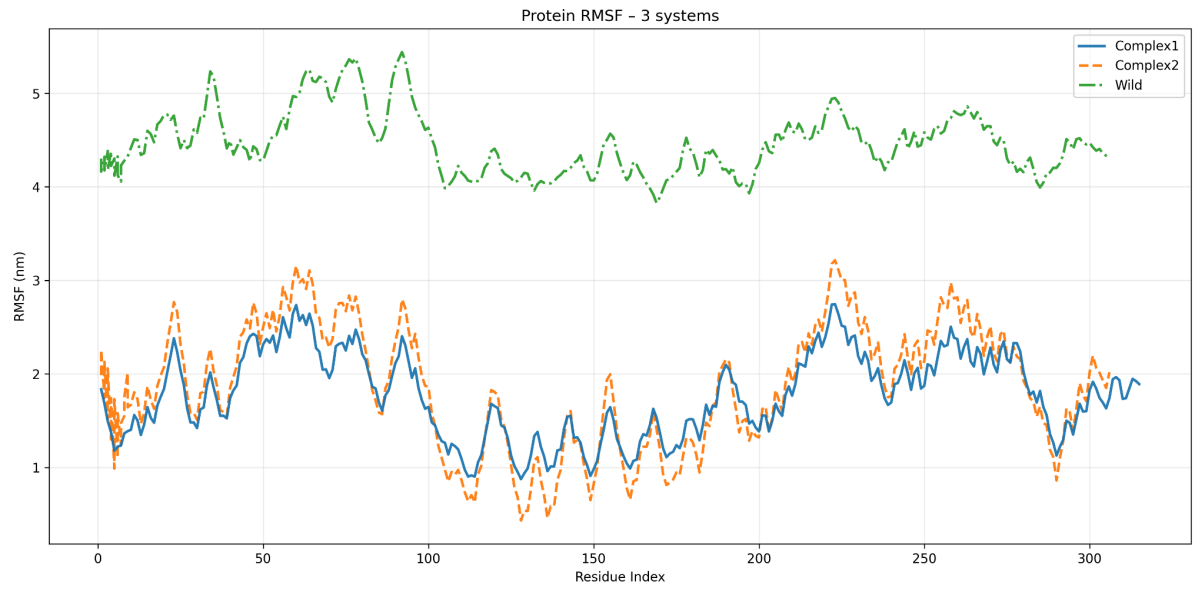
Supplementary material



Supplementary Figure S1. Occupancy obtained by the best peptide of each of the 100 generations.



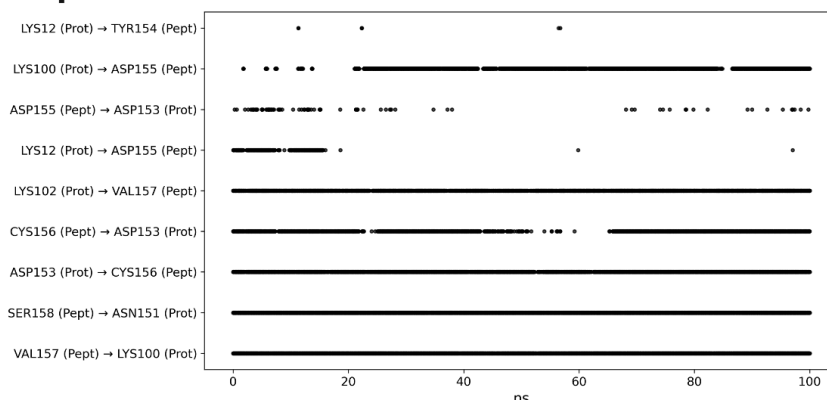
Supplementary Figure S2. RMSD of the three simulated systems – Mpro-N3 (Wild), Mpro-AGVAKAKAV (Complex1), and Mpro-VAKAKAV (Complex2) – over 100 ns of molecular dynamics.



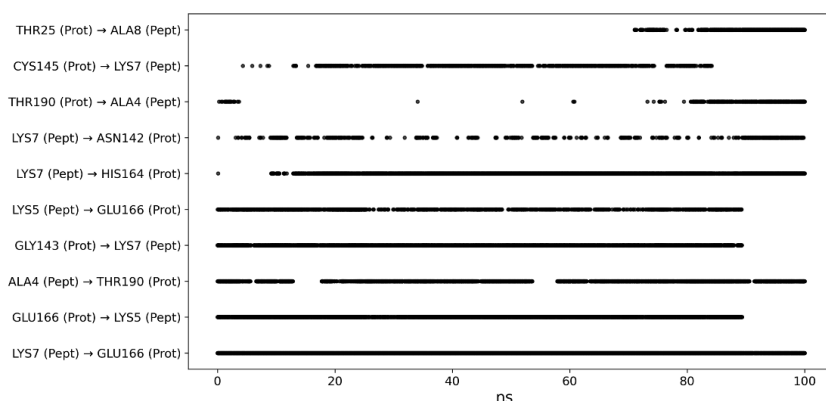
Supplementary Figure S3. RMSF of Mpro in complexes with different ligands – Mpro-N3 (Wild), Complex1 (Mpro-AGVAKAKAV), and Complex2 (Mpro-VAKAKAV) – over 100 ns of molecular dynamics simulation.

Hydrogen Bond Interactions Over Time

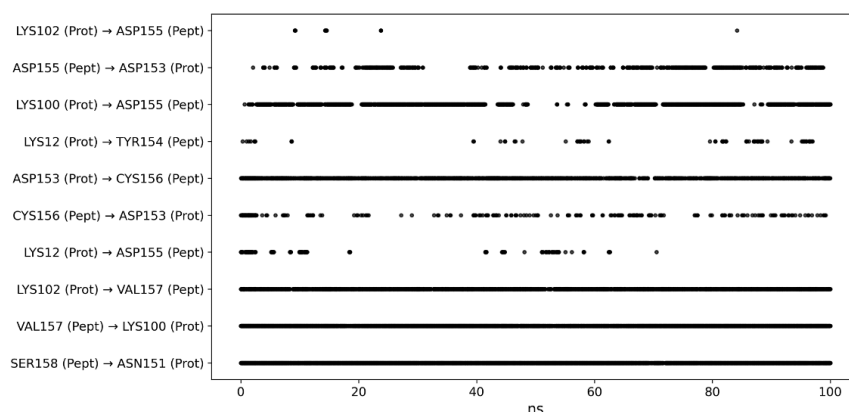
A Mpro-N3



B Mpro-AGVAKAKAV

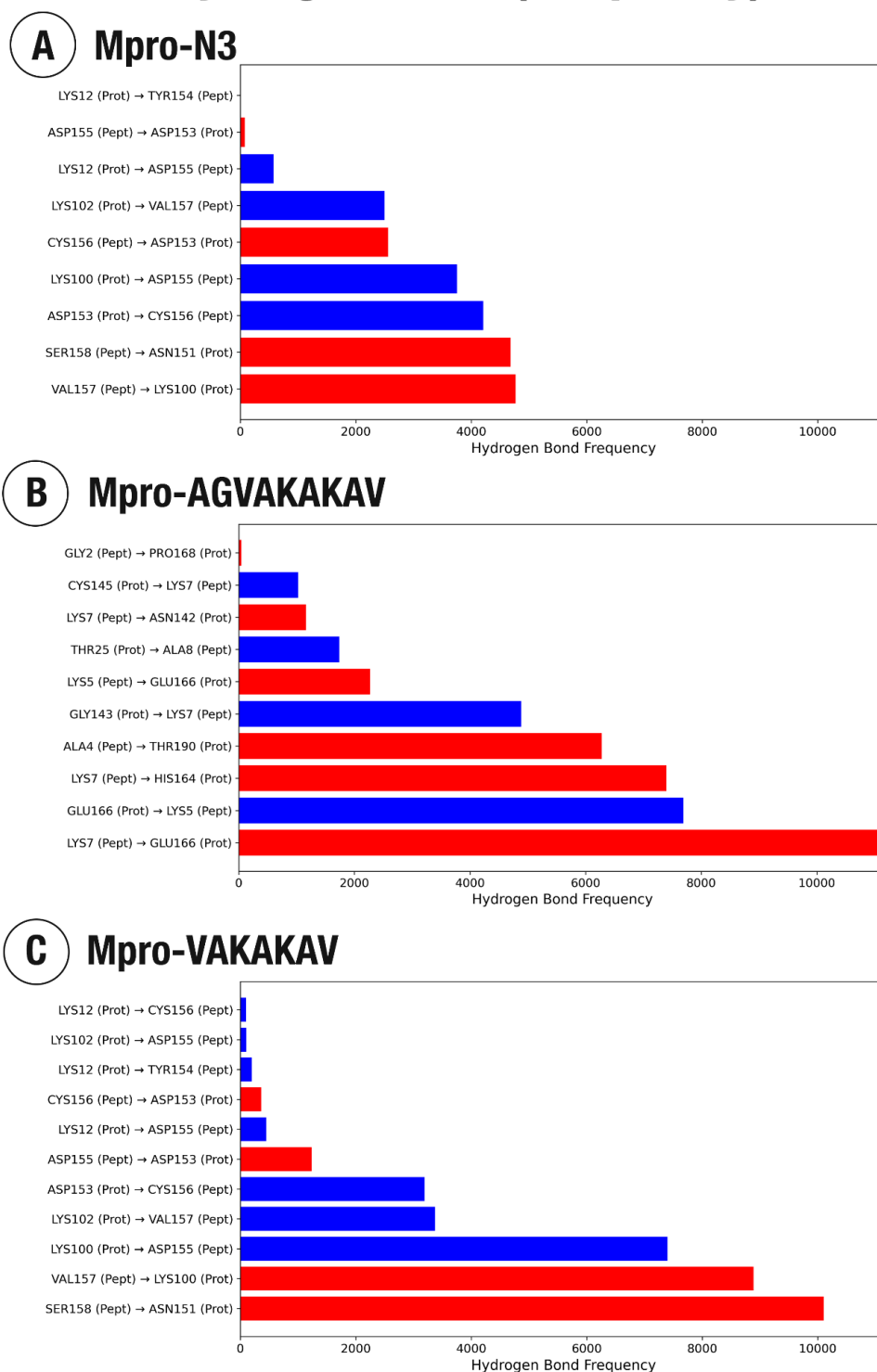


C Mpro-VAKAKAV



Supplementary Figure S4. Hydrogen bonding interactions over 100 ns of molecular dynamics of MPRO with different ligands. Each point represents a frame in which the donor→acceptor pair meets the geometric H-bond criterion (distance H...A ≤ 3.5 Å; angle D-H...A $\geq 135^\circ$). Dense bands indicate persistent (high occupancy) bonds, while gaps indicate intermittent contacts. The labels on the y-axis show the binding direction (Protein → Peptide or Peptide → Protein).

Hydrogen Bond (Frequency)



Supplementary Figure S5. Cumulative hydrogen bond frequencies across 100 ns molecular dynamics simulations of the reference system (Mpro-N3), Mpro-AGVAKAKAV, and Mpro-VAKAKAV. Interactions directed from Protein to Peptide are represented in blue, while those from Peptide to Protein are shown in red.

Physicochemical and Pharmacological Properties

Table S1. Physicochemical properties and predicted toxicity, hemolytic potential, and anti-angiogenic activity of peptides.

Parameter	AGVAKAKAV	VAKAKAV	WWTWTPFHLLV	LTINWQKYFNT
pI ¹	10.02	10.02	7.10	8.94
Prediction ¹	Non-toxic	Non-toxic	Non-toxic	Non-toxic
Prediction ²	Non-Hemolytic HC ₅₀ > 100 µM	Non-Hemolytic HC ₅₀ > 100 µM	Hemolytic	Non-Hemolytic HC ₅₀ > 100 µM
Prediction ³	Anti-angiogenic	Anti-angiogenic	Anti-angiogenic	Anti-angiogenic
Charge ¹	+2	+2	0.5	1
Hydrophobicity ³	0	-0.05	0.23	-0.11
Mol Wt (g/mol) ¹	814.11	685.95	1485.92	1427.79
SVM score ¹	-0.35	-0.23	-0.89	-0.89
Amphipathicity ³	0.82	1.05	0.13	0.45
HC₅₀ (µM) ²	155.883	206.337	90.15	137.787
Observations	Slight loss of amphipathicity	Short and safe cationic profile	Potential cytotoxicity due to high hydrophobicity	High H-bond capacity

pI: isoelectric point (pH where net charge = 0); Charge: theoretical net charge at physiological pH; Hydrophobicity: mean GRAVY score (Kyte–Doolittle scale, dimensionless); Amphipathicity: normalized hydrophobic moment (dimensionless); Mol Wt: theoretical molecular weight (g mol⁻¹).